

Speech Signal Processing

LAB II

Fundamentals of Human Speech Production

DONG Yunyang

dongyy@sustech.edu.cn

411, No. 2, Hui Yuan

Purpose of this lab...

1. Continue to learn MATLAB functionality for speech processing (wave recording, play, etc.)
2. Learn to record & analyse speech (APPAbet symbols, spectrogram, etc.)
3. Learn to use Praat
4. Test the effect of digital filtering on speech perception

Audio Operation

Read or Write Audio

audioread	读取音频文件
audiowrite	写音频文件

```
[audioData,fs] =audioread('mhint_01_01.wav');  
audiowrite('filename.wav', audioData, fs);
```

Object for Audio

audiorecorder	用于录制音频的对象
record	将音频录制到 audiorecorder 对象中
recordblocking	将音频录制到 audiorecorder 对象中，在录制完成前保持控制权
play	从 audiorecorder 对象播放音频
getaudiodata	将录制的音频信号存储在数值数组中

```
recorder = audiorecorder(fs, nBits, nCh);  
disp("Begin speaking for 3s.")  
recorder.recordblocking(3);  
disp("End of recording.")
```

```
recorder.play();
```

```
audioData = recorder.getaudiodata();
```

Praat

- Download Praat
- <http://www.fon.hum.uva.nl/praat/>

		Praat: doing phonetics by computer	
Download Praat: * Macintosh, Windows * Linux, Raspberry Pi, Chromebook * (FreeBSD, SGI, Solaris, HPUX) * license and source code		Information on Praat: * Introductory tutorial: choose Intro from Praat' s Help menus. * Extensive manuals and tutorials: in Praat' s Help menus. * Beginner' s manuals by others. * Paul Boersma' s publications on algorithms and tutorials.	
 Paul		The authors: Paul Boersma and David Weenink Phonetic Sciences , University of Amsterdam visiting: Spuistraat 134 mail: P.O. Box 1642, 1000BP Amsterdam The Netherlands	
Questions, problems, solutions: 1. Many problems can be solved by upgrading to version 6.4.06 of Praat. 2. Make sure you have read the Intro from Praat' s Help menu. 3. If that does not help, use the Search button in Praat' s manual window. 4. Or consult the Frequently Asked Questions directly. 5. There is a user group on the Internet: the Praat User List. 6. If none of the above helps, you can send email to paul.boersma@uva.nl.			

Problem 1

3.18. (MATLAB Exercise) The following MATLAB exercise provides some experience with the process of “segmentation and labeling” of speech. Use the speech file `s5.wav` and its waveform display to do the following:

1. Write out a phonetic transcription of the sentence “Oak is strong and also gives shade” using the ARPAbet symbols.
2. Using the MATLAB functions for plotting and listening to sounds, segment and label the waveform `s5.wav` into regions for each of the sounds in the sentence (including potential regions of silence at the beginning of the speech, and the end of the speech, and even possibly during the speech).
3. Zero out all the vowel regions of the sentence (replace the speech waveform with zero-valued samples during the vowel regions) and listen to the sentence.
4. Zero out all the consonant regions of the sentence (replace the speech waveform with zero-valued samples during the consonant regions) and listen to the sentence.
5. Determine which of the two modified waveforms (i.e., missing the vowels or missing the consonants) is the most “intelligible.” See if someone who had never heard the sentence is able to correctly identify all of the words from either version of the modified waveform.

Oak is strong and also gives shade.

__k__s str_ng __nd __ls__g_v__s sh_d__.

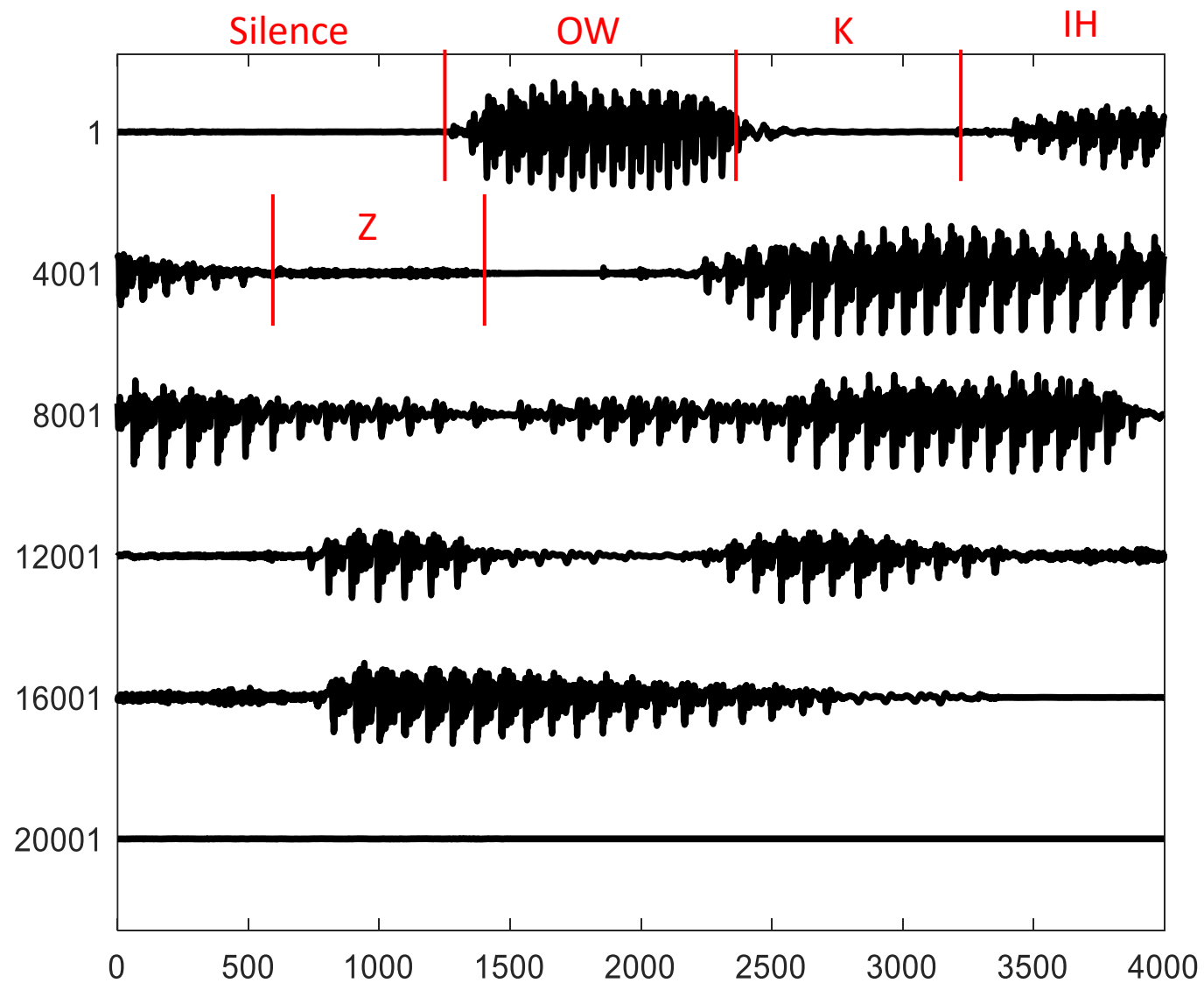
Oa_i__o__a__a__o_i_e__a_e.

Problem 1

TABLE 3.3 Condensed list of phonetic symbols for American English.

IPA			IPA		
Phoneme	ARPAbet	Example	Phoneme	ARPAbet	Example
/i/	IY	be <u>et</u>	/ɨ/	NX	si <u>ng</u>
/I/	IH	bi <u>t</u>	/p/	P	pa <u>t</u>
/ə/	AXR	bu <u>tt</u> er	/t/	T	te <u>n</u>
/ɛ/, /e/	EH	be <u>t</u>	/k/	K	ki <u>t</u>
/æ/	AE	ba <u>t</u>	/b/	B	be <u>t</u>
/a/	AA	Bo <u>b</u>	/d/	D	de <u>bt</u>
/ʌ/	AH	bu <u>t</u>	/g/	G	ge <u>t</u>
/ɔ/	AO	bo <u>u</u> ght	/h/	HH	ha <u>t</u>
/o/	OW	bo <u>a</u> t	/f/	F	fa <u>t</u>
/ʊ/	UH	bo <u>o</u> k	/θ/	TH	thi <u>ng</u>
/u/	UW	bo <u>o</u> t	/s/	S	sa <u>t</u>
/ə/	AX	a <u>b</u> out	/ʃ/, /sh/, /š/	SH	sh <u>ut</u>
/ɪ/	IX	ro <u>s</u> es	/v/	V	va <u>t</u>
/ɜ/	ER	bi <u>r</u> d	/ð/	DH	tha <u>t</u>
/eʏ/	EY	ba <u>i</u> t	/z/	Z	zo <u>o</u>
/aʱ/	AW	do <u>w</u> n	/ʒ/, /zh/, /ž/	ZH	az <u>u</u> re
/aʏ/	AY	bu <u>y</u>	/tʃ/, /č/	CH	chu <u>r</u> ch
/ɔʏ/	OY	bo <u>y</u>	/dʒ/, /j/	JH	ju <u>d</u> ge
/y/	Y	yo <u>u</u>	/ʍ/	WH	whi <u>c</u> h
/w/	W	wi <u>t</u>	/l/	EL	ba <u>tt</u> le
/r/	R	re <u>n</u> t	/m/	EM	bot <u>to</u> m
/l/	L	le <u>t</u>	/n/	EN	bu <u>tt</u> on
/m/	M	me <u>t</u>	/ɾ/	DX	ba <u>tt</u> er
/n/	N	ne <u>t</u>	/ʔ/	Q	(glottal stop)

Problem 1



Problem 2

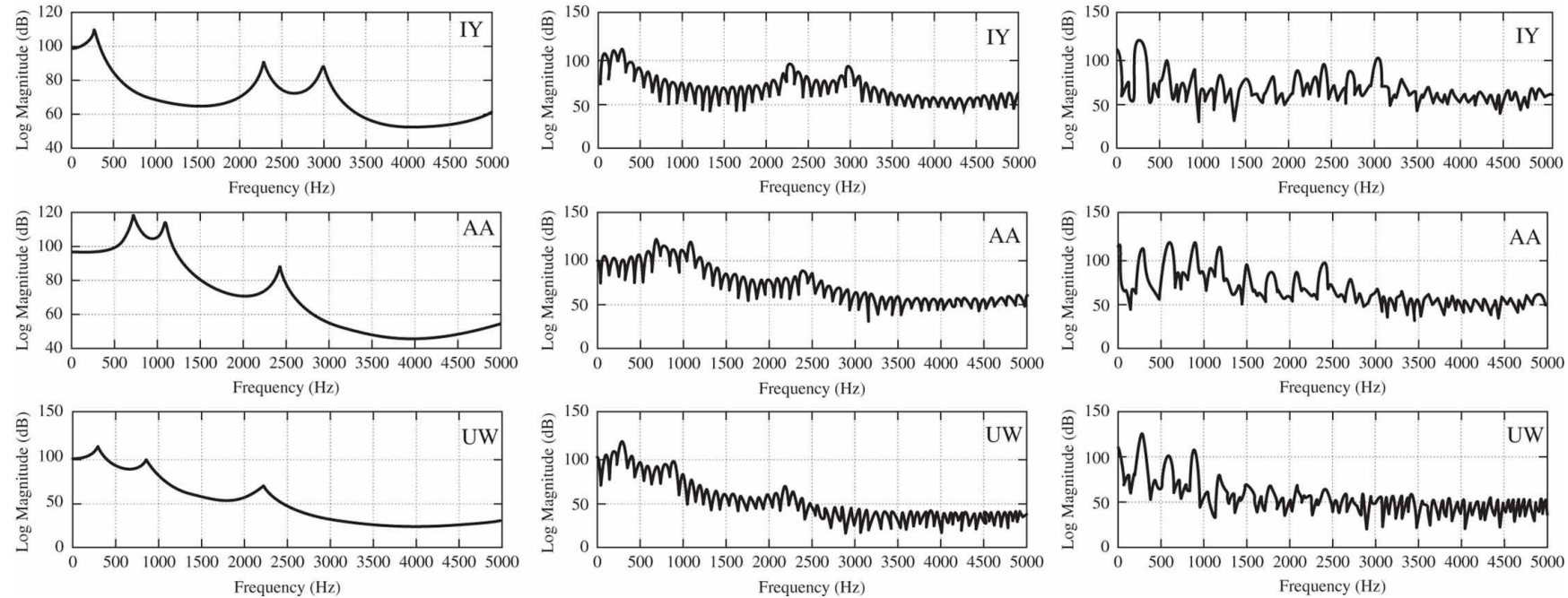
- Under MATLAB environment, use MATLAB codes audiowrite and audiorecorder to record vowels /a/ /i/ /u/.
- Use FFT to analyse the spectrum of the recorded vowels.
- Plot the spectra (amplitude in dB scale), and find the frequencies for **peaks** of spectral envelope.
- Try to use Praat to repeat the above recording and analysis (show the results from Praat)
 - Compare the **formant frequencies** extracted from FFT and Praat analyses

Problem 2

TABLE 3.4 Average formant frequencies for the vowels of American English. (After Peterson and Barney [287]. © [1952] Acoustical Society of America.)

Average Vowel Formant Frequencies					
ARPAbet Symbol	IPA Symbol	Typical Word	F_1	F_2	F_3
IY	i	(beet)	270	2290	3010
IH	ɪ	(bit)	390	1990	2550
EH	ɛ	(bet)	530	1840	2480
AE	æ	(bat)	660	1720	2410
AH	ʌ	(but)	520	1190	2390
AA	a	(hot)	730	1090	2440
AO	ɔ	(bought)	570	840	2410
ER	ɜ	(bird)	490	1350	1690
UH	U	(foot)	440	1020	2240
UW	u	(boot)	300	870	2240

Problem 2



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Figure 3.22 Canonic vowel spectra for the three vowels /iY/, /AA/, and /UW/ at the corners of the vowel triangle; left pane shows the vocal tract frequency response; middle pane shows the spectra with a 100 Hz pitch frequency excitation; right pane shows the spectra with a 300 Hz pitch frequency excitation.

Problem 3

test_16k.wav

- 3.21.** (MATLAB Exercise) Write a MATLAB program to highpass filter a speech file in order to eliminate any potential DC offset and/or 60 Hz hum that may have been recorded with the speech file. The first thing you must do for this exercise is design a digital highpass filter with appropriate cutoff frequencies (edge of stopband and edge of passband) so that the 60 Hz component is attenuated by at least 40 dB. (Hint: You will find that a linear phase FIR filter probably works best for this exercise.) Once the filter has been designed, filter the speech signal and ~~overwrite the original file~~ with the highpass filtered speech.

create a new file